

Planck and Bohm the Men Whose Fate Is “Uncomputed”

—Part 6: The Ultimate Reality—

Characters:

- **The Objective Observer (Salviati):** One who possesses deep insights into Digital Signal Theory and the discrete universe.
- **The Existing Intellect (Simplicio):** One who reveres Quantum Mechanics as a "sublime truth."
- **The Seeker (Sagredo):** One who seeks to unravel the universe's specifications through simple logic.

Act I: The Formidable Complex Number

Sagredo: By the way, we spoke earlier of complex numbers—how the phase information they allow us to express is the very foundation of quantum mechanics. But that isn't the only information a complex number holds. A complex number, by its very nature, possesses information regarding its "direction of rotation." I have realized that this information is what divides reality itself.

Act II: Our Universe

Simplicio: Direction of rotation? What could that possibly mean in a physical sense?

Sagredo: In our world, the *curl* operator signifies a right-handed rotation. This can be rephrased as our world having the property of being represented by complex basis functions with a right-handed rotational direction—that is, positive frequency components.

Act III: The Alias of Our Universe

Simplicio: Then what of a world represented by complex basis functions with negative frequency components?

Sagredo: It is a world that necessitates the left-handed curl operator. In our world, governed by the right-handed curl operator, a positive charge is subject to a right-handed force; this is what we define as "matter." Conversely, in a world where the left-handed curl operator is required, a positive charge is subject to a left-handed force—and this is the true definition of "antimatter." Physicists, reluctant to define a left-handed curl operator, introduced the concepts of negative velocity and time proceeding toward negative infinity, describing it as "antimatter appearing to move backward in time." However, this is not an essential interpretation; it is merely an engineering technique. What matters in physics is that, if identical results are obtained, one must adopt the logic that preserves symmetry and causality, even if the mathematical formulas become more complex. This is the fundamental difference between the Copenhagen interpretation and the Bohmian interpretation [3][4]. It goes without saying which of the two these theories correspond to.

Simplicio: ……

Salviati: Correct. Our physical laws adopt the "right-hand rule" simply because our universe selected right-handed rotation as its fundamental operation. Since mathematics permits both right-handed and left-handed rotation, the Architect's computing engine outputs both as "reality."

Act IV: The Ultimate Truth

Sagredo: The ultimate truth revealed in this *New Dialogue* is the universal specification that "whatever exists mathematically, exists in reality." One might even call it the specification of the Architect's computing engine.

Sagredo: Precisely. The entire stage was a **Four-Dimensional Digital Matrix Memory Space** where computation and self-recording are repeated. And "computation" was the transition of phase information—it was "time" itself. The universe was not playing dice. What we perceived as "probability" was merely the "**quantization error**" that inevitably occurs when outputting computational results onto a digital matrix with a lattice size of $\$h\$$.

(Curtain Falls)

Notes:

1. **Alias:** In the direction of rotation, there is no "real" or "fake." An "alias" is nothing more than a product of the ego's subjectivity claiming itself as the original. Aliases are twins—mathematically, they are N-tuplets. Here, one should recall the specification: "Whatever exists mathematically, exists in reality."
2. **Four-Dimensional Digital Matrix Memory Space:** The original computing engine of the Architect is different from our own degraded devices. It requires neither registers nor data buses. Within the main memory space, all computation and self-recording are repeated.

【References】

- [3] Bohm, D. (1952). "A Suggested Interpretation of the Quantum Theory in Terms of 'Hidden' Variables.". *Physical Review*, 85(2), 166-193.
- [4] Takeuchi, K. (2022). *Zero kara Manabu Ryoshirikyaku Fukyu-ban* [Learning Quantum Mechanics from Scratch, Popular Edition]. Kodansha. (in Japanese).

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